Important note:

Notice that the market impact formulation for a one-period trade schedule reduces to:

$$MI_{bp} = \hat{b}_1 \cdot I^* \cdot \left(\frac{X}{V_t}\right) + (1 - \hat{b}_1) \cdot I^* \tag{7.16}$$

This is the same formulation as Equation 7.15 with $a_4 = 1$. The importance of this equation is that it will be used to calibrate the market impact parameters for the trade schedule solution (shown below). Recall that this was also the simplified version of the model described in the two-step regression process shown in Chapter 5, Estimating I-Star Model Parameters.

Market impact cost across stock is an additive function. Therefore, the impact for a basket of stock is the sum of impacts for the entire basket. The addition problem is simplified when market impact is expressed in dollar units so that we do not need to worry about trade value weightings across stocks. These are:

Market Impact for a Basket of Stock

$$MI_{\$}(POV) = \sum_{i=1}^{m} (\hat{b}_1 \cdot I_i^* \cdot POV_i^{\hat{a}_4} + (1 - \hat{b}_1) \cdot I_i^*) \tag{7.17}$$

$$MI_{\$}(\alpha) = \sum_{i=1}^{m} (\hat{b}_1 \cdot I_i^* \cdot \alpha_i^{\hat{a}_4} + (1 - \hat{b}_1) \cdot I_i^*) \tag{7.18}$$

$$MI_{\$}(x_k) = \sum_{i=1}^{m} \sum_{k=1}^{n} \left(b_1 \cdot I_i^* \cdot \frac{x_{ik}^2}{X_i \cdot v_{ik}} \right) + (1 - b_1) \cdot I_i^* \tag{7.19}$$

TIMING RISK EQUATION

The timing risk for an order executed over a period of time t^* following a constant trading strategy is as follows:

$$TR(t^*)_{bp} = \sigma \cdot \sqrt{\frac{1}{250} \cdot \frac{1}{3} \cdot t^*} \cdot 10^4 bp \tag{7.20}$$

This equation simply scales price volatility for the corresponding trading period t^* and adjusts for the trade strategy (e.g., decreasing portfolio size). For example, σ is first scaled to a one-day period by dividing by $\sqrt{250}$, then this quantity is scaled to the appropriate trading time period